FOR A FLAT MORPHISM COHEN-MACAULAYNESS AT A POINT IS EQUIVALENT TO THE SHRIEK OF THE STRUCTURE SHEAF HAVING ONE COHOMOLOGY SHEAF NEARBY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0C0Z (sources/stacks-duality/pinned-original-source.tex line 6284).

Allowed standard prerequisites (editorial).

- The Cohen-Macaulay criterion through the shriek of the structure sheaf. Same chapter, Lemma labelled lemma-CM-shriek, cited at proof lines 6295-6297 and 6304-6306. Gives one direction and the equality of the component dimensions.
- Openness of the Cohen-Macaulay locus for a flat finite presentation morphism and boundedness of fibre dimensions. More on Morphisms, Lemma labelled more-morphisms-lemma-flat-finite-presentation-CM-open, cited at proof lines 6304-6306, and Morphisms, Lemma labelled morphisms-lemma-openness-bounded-dimension-fibres, cited at proof lines 6308-6310. Show the generic points are Cohen-Macaulay and bound the fibre dimensions.
- Flatness of the single cohomology sheaf and the pullback along the fibre. Same chapter, Lemma labelled lemma-flat-shriek, cited at proof lines 6310-6312, and Derived Categories of Schemes, Lemma labelled perfect-lemma-tor-independence-and-tor-amplitude, cited at proof lines 6314-6316. Compute the pullback of the shriek to the fibre.
- The dualizing complex of a fibre and the Cohen-Macaulay criterion for it. Same chapter, Lemmas labelled lemma-relative-dualizing-fibres and lemma-dualizing-module-CM-scheme, cited at proof lines 6316-6319. Identify the pullback as a dualizing complex and conclude.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BV8: sources/stacks-duality/pinned-original-source.tex, source lines 5978–5978. Same-chapter pointer into the author environment lemma at source line 5978; invoked at proof line 6296 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 045U: sources/stacks-chow/tags.txt, source lines 5449–5449. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6303 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02FZ: sources/stacks-chow/tags.txt, source lines 3210–3210. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6307 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BV6: sources/stacks-duality/pinned-original-source.tex, source lines 5915–5915. Same-chapter pointer into the author environment lemma at source line 5915; invoked at proof line 6309 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0C0V: sources/stacks-chow/tags.txt, source lines 15100–15100. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6313 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BZZ: sources/stacks-duality/pinned-original-source.tex, source lines 5130–5130. Same-chapter pointer into the author environment lemma at source line 5130; invoked at proof line 6315 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AWT: sources/stacks-duality/pinned-original-source.tex, source lines 6214–6214. Same-chapter pointer into the author environment lemma at source line 6214; invoked at proof line 6317 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0F42 let $f : X \to Y$ be a morphism of FTS_S. Let $x \in X$. If f is flat, then the following are equivalent*

(1) *f is Cohen-Macaulay at x,*
(2) *$f^!\mathcal{O}_Y$ has a unique nonzero cohomology sheaf in a neighbourhood of x.*

Proof. One direction of the lemma follows from Lemma 0BV8. To prove the converse, we may assume $f^!\mathcal{O}_Y$ has a unique nonzero cohomology sheaf. Let $y = f(x)$. Let $\xi_1, \ldots, \xi_n \in X_y$ be the generic points of the fibre X_y specializing to x. Let $d_1, \ldots, d_n$ be the dimensions of the corresponding irreducible components of X_y.

The morphism $f : X \to Y$ is Cohen-Macaulay at η_i by More on Morphisms, Lemma 045U. Hence by Lemma 0BV8 we see that $d_1 = \ldots = d_n$. If d denotes the common value, then $d = \dim_x(X_y)$. After shrinking X we may assume all fibres have dimension at most d (Morphisms, Lemma 02FZ). Then the only nonzero cohomology sheaf $\omega = H^{-d}(f^!\mathcal{O}_Y)$ is flat over Y by Lemma 0BV6. Hence, if $h : X_y \to X$ denotes the canonical morphism, then $Lh^*(f^!\mathcal{O}_Y) = Lh^*(\omega[d]) = (h^*\omega)[d]$ by Derived Categories of Schemes, Lemma 0C0V. Thus $h^*\omega[d]$ is the dualizing complex of X_y by Lemma 0BZZ. Hence X_y is Cohen-Macaulay by Lemma 0AWT. This proves f is Cohen-Macaulay at x as desired. □